Recursive Clock Tracking for Photon-Counting Quantum Key Distribution: Kalman Synchronization, Error and Lock-Loss Analysis, and Finite-Key Performance

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***Abstract*—Single-photon detectors in quantum key distribution (QKD) are gated for about 2 ns, so the receiver must know the arrival time of every photon to a small fraction of a nanosecond. Recent work recovers the clock directly from the quantum signal, either by fitting batches of detection times or by comparing histograms of consecutive acquisitions, but these estimators discard what they have learned at every correction and degrade when detections are rare. We treat clock recovery for photon-counting QKD as a state-estimation problem with sparse, randomly timed, outlier-contaminated measurements. We propose a Kalman tracker of the timing offset and drift that is updated after every detection, rejects dark counts with an innovation gate, and runs either on a small share of pilot pulses or directly on the ordinary key detections without any extra photon. We derive the steady-state timing error, the probability of losing lock, and a minimum-pilot design rule, and check them against Monte Carlo simulation. In a decoy-state BB84 link with composable finite-key bounds, simulated with 10^{11}–3×10^{14} pulses per figure, Kalman tracking on key detections lowers the RMS gate error by a factor of 3–5 relative to a histogram method in the style of Millet *et al.* and by more than 10 relative to batch re-synchronization, from 25 to 125 km, and it is the only method that still retains a substantial fraction of the ideal key rate at 125 km, although the absolute rate there is small. A classical timing reference matches the tracker only if its residual error is below about 0.2 ns; in addition, a delay of such a reference can leak up to 0.45 bit per sifted bit under detector mismatch, which in-band synchronization absorbs. All results are from simulation, and the assumptions are stated.**

***Index Terms*—Quantum key distribution, clock synchronization, Kalman filter, finite-key analysis, decoy states, time-shift attack.**

I. Introduction

QUANTUM key distribution (QKD) lets two parties establish a secret key whose security rests on physical law, and the BB84 protocol [1] with weak coherent pulses and decoy states [2]–[4] is the basis of most deployed systems. A practical receiver opens a detector gate of a couple of nanoseconds around each expected arrival. If the receiver clock is offset by a few nanoseconds, drifts by parts per million, or wanders, the photons fall outside the gate, the key rate collapses, and the share of dark counts in the detections rises. Clock synchronization is therefore as much a part of the link budget as fibre loss.

The conventional solution is a separate timing channel, such as a bright synchronization laser or a White Rabbit link [5]. It needs extra hardware and fibre, and it adds an attack surface (Sec. V). A growing body of work instead recovers the clock from the quantum signal itself. Qubit4Sync [6] correlates the detection times with a known qubit sequence, Cochran and Gauthier [7] use Bayesian batch estimation, and further variants and field trials have been reported [8]–[12]. The most recent proposal, by Millet *et al.* [13], estimates drift and offset from the circular means of two consecutive timing histograms of the key detections, needs no extra photons, and was validated on commercial hardware. All of these estimators work in batches: a correction is computed from a window of detections and applied, and the next window starts afresh. They cannot weigh old against new information, and the estimate is frozen between corrections, which is costly when detections are rare (a few pilot detections per millisecond at 100 km) or the oscillator wanders.

Kalman filtering is a standard tool for exactly this situation. In QKD it has been applied to track the reference phase and polarization of continuous-variable systems from bright pilot signals [14]–[18], and in classical networks it underlies clock servos for IEEE 1588 and wireless sensor networks [19]–[21]. To our knowledge it has not been applied to clock tracking from sparse single-photon detections in photon-counting QKD, where the measurements are Poisson-timed clicks, a fraction of which are dark-count outliers, and where a lost lock is catastrophic. That is the gap addressed here.

The contributions of this paper are:

1) A two-stage clock estimator for photon-counting QKD: drift-search acquisition on a short preamble, followed by a Kalman tracker of offset and drift that is updated after every detection and rejects dark counts with an innovation gate. It runs on pilot pulses or, with no extra photons, on the key detections (Sec. III).
2) An analytical model of the tracker: the steady-state timing error, the probability of losing lock, and a minimum pilot overhead as a function of distance and oscillator quality, validated against simulation (Sec. IV).
3) A benchmark of eight synchronization strategies under composable decoy-state finite-key bounds, in which the two published estimator families are tuned to their best settings (Sec. VI).
4) A timing-layer security analysis: the leakage caused by delaying a classical reference, the absorption of such a delay by in-band synchronization, a variance test for

Manuscript draft. Numerical results are simulation based.

TABLE I
DEFAULT SIMULATION PARAMETERS.

Parameter	Value
Slot period T	10 ns
Fibre loss, detector efficiency	0.2 dB/km, 0.6
Dark counts, gate width W	10^4 Hz per detector, 2 ns
Arrival jitter σ	0.35 ns (Gaussian)
Initial offset, drift	4 ns, 20 ppm
Oscillator wander w	0.01 (stable), 0.2 (poor) ppm/$\sqrt{\text{ms}}$
Intensities μ_1, μ_2, μ_3	0.5, 0.2, 0.0002
Intensity probabilities, p_Z	0.6, 0.2, 0.2; 0.8
Channel bit error	2% (Pauli $p = 0.03$)
f_{EC}, ε_{sec}, ε_{cor}	1.16, 10^{-10}, 10^{-15}
Key block, timing segment	10^{10} slots (100 s), 10^8 slots (1 s)
Pilot share, classical residual	1%, 0.5 ns (rms)
Preamble	64 expected detections, spacing 5 slots

random time shifts, and a bound on the leakage of synchronizing on key detections (Sec. V).

II. System Model

Figure 1 shows the link. Table I lists the default parameters, which are used throughout unless stated.

A. Source, channel and detection

Alice emits one pulse per slot with intensity μ_k chosen with probability p_k and encodes it in the Z basis with probability p_Z and the X basis otherwise; Bob chooses his basis in the same way. Let $T_{\text{ch}} = 10^{-\alpha L/10}$ be the fibre transmittance for loss α and length L, and $\eta = \eta_d T_{\text{ch}}$ the total efficiency with detector efficiency η_d. If the photon falls inside the gate with probability c (the timing capture, below) and the dark-click probability per gate is $p_d = 1 - e^{-2R_d W}$ for dark rate R_d and two detectors, a slot with intensity μ yields a click with probability and an error with probability

$$p_{\text{ph}}(\mu) = 1 - e^{-\eta c \mu}, \qquad p_{\text{click}}(\mu) = p_{\text{ph}} + (1 - p_{\text{ph}})\, p_d, \tag{1}$$

$$p_{\text{err}}(\mu) = \big[p_{\text{ph}}(1 - p_d)\, e_{\text{sig}} + \tfrac{1}{2} p_d\big] / p_{\text{click}}(\mu). \tag{2}$$

Here e_{sig} is the bit error of a genuine click, combining channel noise e_{ch} and an intercept-resend fraction f of a spy as $e_{\text{ch}}(1 - f/4) + (f/4)(1 - e_{\text{ch}})$. Poor timing lowers c, which reduces the genuine clicks while leaving the dark clicks unchanged and therefore raises the error.

B. Timing layer

Relative to Bob's nominal gate schedule, a photon sent at time t arrives with the timing error

$$e(t) = e_0 + \nu_0 t + w(t) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2), \tag{3}$$

where e_0 is the initial offset, ν_0 the fractional frequency offset (drift), $w(t)$ the integrated random-walk frequency noise of Bob's oscillator, and ε the arrival jitter. The random walk is characterized by a diffusion constant $q = (w \cdot 10^{-6})^2 / 10^6$ ns^{-1} for a wander w in ppm/$\sqrt{\text{ms}}$. Bob opens a gate of width W at his estimate $\hat{e}(t)$. With gate error $d(t) = e(t) - \varepsilon - \hat{e}(t)$, the capture probability is

$$c(d) = \Phi\!\left(\frac{W/2 - d}{\sigma}\right) - \Phi\!\left(\frac{-W/2 - d}{\sigma}\right), \tag{4}$$

with Φ the standard normal distribution function. For $W = 2$ ns and $\sigma = 0.35$ ns, a perfect clock gives $c = 99.57\%$, a gate error of 0.5 ns gives 92.4%, 1 ns gives 50%, and 2 ns gives 0.2%. We average c over the key part of the timing segment and call this the timing capture. Synchronization determines $\hat{e}(t)$; its quality is measured by the RMS gate error and the capture.

C. Finite-key rate

We use the decoy-state finite-key bound of Lim *et al.* [4] with three intensities. From the Z detections n_Z and errors m_Z and the X detections n_X and errors m_X at each intensity, widened by Hoeffding terms $\delta = \sqrt{(n/2)\ln(21/\varepsilon_{\text{sec}})}$, one bounds the number s_0 of vacuum and s_1 of single-photon Z detections and the single-photon phase error φ. The secret key length is

$$\ell = s_0 + s_1\big[1 - h(\varphi)\big] - \lambda_{\text{EC}} - 6\log_2\frac{21}{\varepsilon_{\text{sec}}} - \log_2\frac{2}{\varepsilon_{\text{cor}}}, \tag{5}$$

with h the binary entropy, $\lambda_{\text{EC}} = f_{\text{EC}}\, n_Z\, h(e_Z)$ the error-correction leakage, and a block is rejected if $\ell \le 0$. The key rate is ℓ divided by the number of slots of the key block, in bits per slot (multiply by the pulse rate for bits per second). Slots spent on preamble and pilots are removed from the key slots, so synchronization is paid for in the key rate. The asymptotic rate, computed from the same counts without the margins, and the PLOB bound [22] serve as references.

D. Two time scales

Timing is simulated pulse by pulse over a segment of 10^8 slots (1 s), including the preamble, the pilots or key detections, the dark counts and the estimator. The resulting timing capture is then applied to the whole key block of 10^{10} slots (100 s), whose detection counts are drawn exactly from binomial and multinomial distributions. This assumes that the synchronization is statistically stationary after the first second, which holds for a locked tracker; it is listed among the limitations in Sec. VII.

III. Synchronization Strategies

All strategies return an estimate $\hat{e}(t)$. We compare eight: no synchronization ($\hat{e} = 0$); a classical reference (a constant residual drawn from $\mathcal{N}(0, 0.5^2)$ ns per block, with no photons used); preamble-only QCS (stage 1 below, extrapolated); batch re-synchronization; histogram re-synchronization; Kalman tracking on pilots; Kalman tracking on key detections; and perfect synchronization ($\hat{e} = e$).

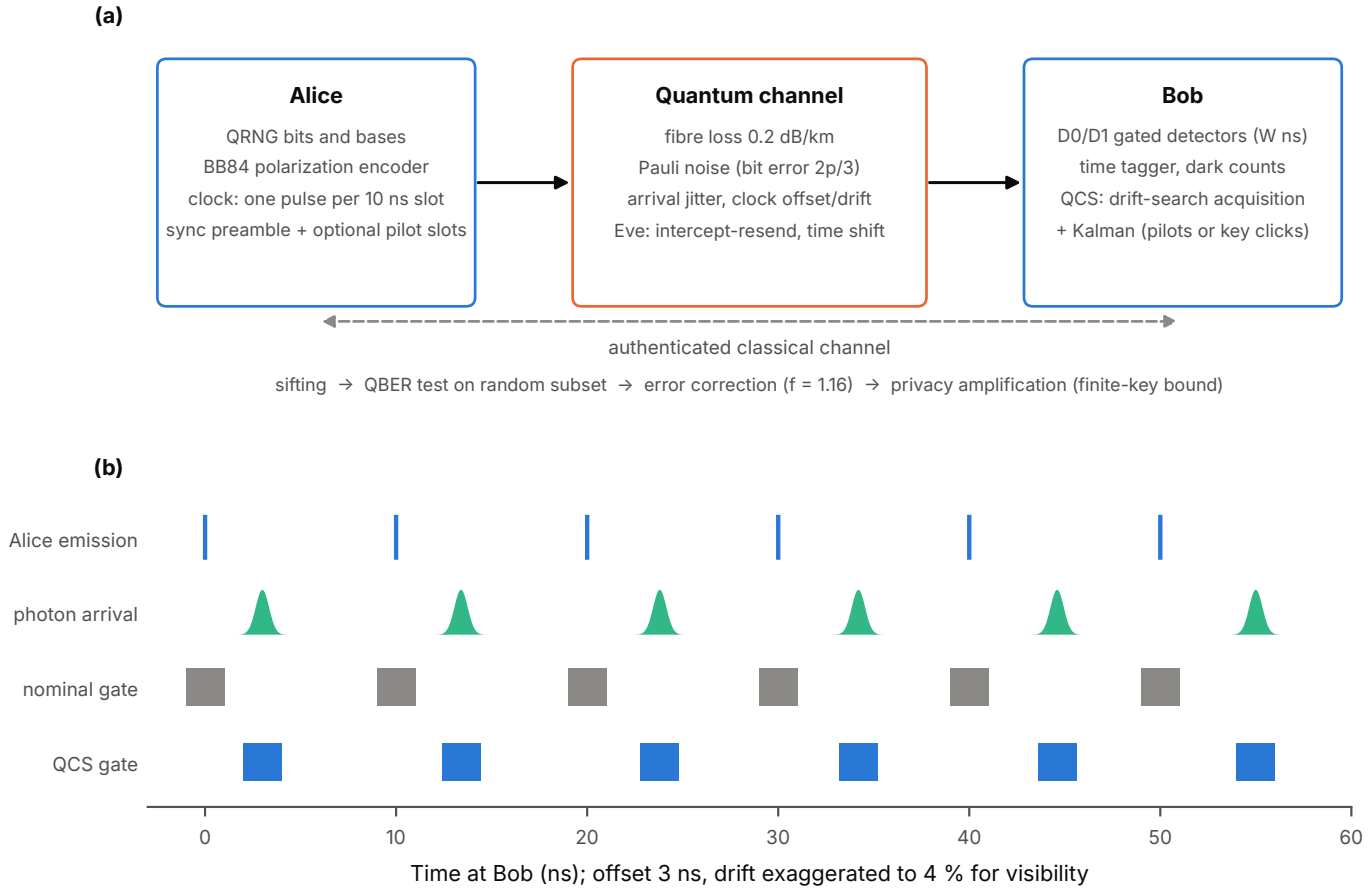


Fig. 1. System model. (a) Alice, the fibre channel and Bob, with the synchronization loop and the classical post-processing chain. (b) Timing diagram: photon arrivals are shifted by clock offset and drift, so the nominal gates miss them, while the synchronized gates catch them.

A. Acquisition (stage 1, common to all QCS methods)

Alice sends a preamble of one sync pulse every 5 slots (50 ns) until about 64 detections are expected. Bob time-tags every click. Each tag is known only modulo the preamble spacing, since Bob does not know which pulse produced it, and dark counts are uniform over the spacing. For every candidate drift ν on a grid over ± 100 ppm, the tags are de-rotated by νt, wrapped, and histogrammed in bins of width 2σ; the ν with the tallest peak is selected. Tags within 3σ of the peak are unwrapped and a least-squares line gives the offset, the drift and their covariance (a 2×2 matrix) at the last tag. The accuracy of this stage is compared with the Cramér–Rao bound in Sec. IV. Preamble-only QCS extrapolates this line for the whole block; with a drift known to $\sim 10^{-7}$ the gate is lost within milliseconds.

B. Batch re-synchronization (prior art)

Representative of Qubit4Sync-type methods [6], [7], and run on pilot pulses (one per 100 slots). Bob predicts each pilot click from the current line, records the residual of the clicks that fall within $\pm T/2$, and after 32 clicks repeats the drift-search fit of stage 1 on the residuals (± 100 ppm), adds the result to the line and holds it until the next batch is complete.

C. Histogram re-synchronization (after Millet et al.)

Following [13], which uses the key detections and no pilots, the residuals of the detection times are folded onto the slot period T and mapped to phases $\theta = 2\pi r/T$. Two consecutive histograms of duration T_{int} give circular means $m_1 = \mathbb{E}[e^{i\theta}]$ and m_2. The drift and offset corrections are

$$\hat{\nu} = \frac{T}{2\pi T_{\mathrm{int}}} \arg(m_2 \bar{m}_1), \qquad \hat{\tau} = \frac{T}{2\pi} \arg(m_2) + \hat{\nu} \frac{T_{\mathrm{int}}}{2}, \quad (6)$$

applied once per pair. The windows start at about 10 detections and grow by a factor of 4 after every three corrections up to a maximum duration, as in the paper's start-up schedule. Because the best maximum duration depends on distance and oscillator quality, we simulate 100 μs, 300 μs, 1 ms and 3 ms and keep the best at every point (Sec. VI). This is our reimplementation from the published description, not the authors' code.

D. Kalman tracking (this work)

The state is $\mathbf{x} = [\tau, \nu]^{\top}$ (timing error and drift) with covariance $\mathbf{P}$. During a gap Δ since the last update,

$$\mathbf{x}^{-} = \mathbf{F}\mathbf{x}, \quad \mathbf{F} = \begin{bmatrix} 1 & \Delta \\ 0 & 1 \end{bmatrix}, \quad (7)$$

$$\mathbf{P}^{-} = \mathbf{F}\mathbf{P}\mathbf{F}^{\top} + q \begin{bmatrix} \Delta^3/3 & \Delta^2/2 \\ \Delta^2/2 & \Delta \end{bmatrix}, \quad (8)$$

where the drift performs a random walk of strength q and, with it, the timing error. A detection with time tag $z = \tau + n$, $n \sim \mathcal{N}(0, r)$, $r = \sigma^2$, produces the innovation $y = z - \tau^-$ with variance $S = P_{11}^- + r$. It is rejected as a dark count if $y^2 > \gamma^2 S$ ($\gamma = 3$) and otherwise used:

$$\mathbf{K} = \mathbf{P}^- \mathbf{h}^\top / S, \quad \mathbf{x} = \mathbf{x}^- + \mathbf{K} y, \quad \mathbf{P} = (\mathbf{I} - \mathbf{K}\mathbf{h})\mathbf{P}^-, \tag{9}$$

with $\mathbf{h} = [1, 0]$. The initial state and covariance come from acquisition. The state has two elements, so the cost per detection is a few multiplications. It carries its uncertainty from update to update: after a long silence P_{11}^- is large and the next detection is trusted more, while after many detections the gain falls and each detection moves the estimate only a little. The process noise q is set from the oscillator wander (with a floor corresponding to 0.005 ppm$/\sqrt{\text{ms}}$).

Pilot version. One sync pulse in every 100 slots (1% overhead); detections are processed as they arrive, and the pilot listening window is the slot, $\pm T/2$ around the prediction.

Key-detection version. The same filter runs on the time tags of the key detections (all intensities), with no pilots. The arrival time of a click does not depend on the bit or the basis in an ideal receiver, so every click carries timing information at no cost in photons (security in Sec. V). To keep the computation tractable at short range, up to 10^6 detections per timing segment are used, and they are processed in groups of at most 25: each click is gated against the group prediction, and the accepted tags are averaged into one measurement at their mean time with variance r/k for k accepted clicks. For a linear drift this equals sequential processing; the group spans a few microseconds. The cap makes the short-range accuracy a conservative figure.

IV. Performance Analysis

A. Steady-state timing error

Let detections arrive as a Poisson process with mean gap Δ (for pilots $\Delta = n_p T / p_{\text{sync}}$, with n_p slots per pilot and $p_{\text{sync}} = 1 - e^{-\eta\mu_1}$; for key detections the mean gap between processed clicks). Iterating (8)–(9) with a fixed gap Δ converges to a covariance $\mathbf{P}_\infty$ just after an update (the discrete algebraic Riccati solution). Between updates the error variance at age t is

$$V(t) = P_{11} + 2tP_{12} + t^2 P_{22} + \tfrac{1}{3} q t^3. \tag{10}$$

The age of a random instant is exponentially distributed with mean Δ, because the arrivals are memoryless.

Result 1 (Mean-square gate error)**.** *Using* $\int_0^\infty t^n e^{-t/\Delta} \Delta^{-1} \mathrm{d}t = n!\, \Delta^n$,

$$\text{MS} = \mathbb{E}[V] = P_{11} + 2P_{12}\Delta + 2P_{22}\Delta^2 + 2q\Delta^3, \tag{11}$$

The RMS gate error is $\sqrt{\text{MS}}$. The terms are: the noise left by the last update, the uncertainty of the drift multiplied by the waiting time, and the wander of the oscillator during the wait. The covariance in (11) is the fixed-gap steady state, which approximates the random gaps.

TABLE II
Minimum pilot share (% of slots) for $P_{\text{lost}} \le 1\%$ per second.

Distance	$w = 0.01$	$w = 0.05$	$w = 0.2$
50 km	0.026	0.09	0.27
75 km	0.10	0.34	0.93
100 km	0.37	1.2	3.2
125 km	1.3	4.2	11
150 km	4.5	14	50

B. Probability of losing lock

Two mechanisms are modelled as independent hazards. *Overrun:* if the error leaves the pilot window $\pm h$, $h = T/2$, during a gap g, the next pilot is missed and the tracker cannot recover. The miss probability for a gap g is $2[1 - \Phi(h/\sqrt{V(g)})]$; averaging over exponential gaps gives p_{gap} and the rate $\lambda_{\text{over}} = -\ln(1 - p_{\text{gap}})/\Delta$. *Dark capture:* a dark click inside the innovation gate is accepted, and is harmful when the prediction is less certain than a single click ($V > r$). With p'_d the dark-click probability per sync slot of period T_p and $a(t) = \min\big(1, \gamma\sqrt{V(t) + r}/h\big)$ the fraction of the window accepted by the gate,

$$\lambda_{\text{dark}} = \frac{p'_d}{T_p} \int_0^\infty a(t)\, \mathbf{1}[V(t) > r]\, \frac{e^{-t/\Delta}}{\Delta}\, \mathrm{d}t. \tag{12}$$

The probability of losing lock within a segment of duration T_{seg} is

$$P_{\text{lost}} = 1 - \exp\big[-(\lambda_{\text{over}} + \lambda_{\text{dark}})\, T_{\text{seg}}\big]. \tag{13}$$

These formulas are for the pilot tracker. For the key-detection version we give only the error (11); no lock loss was observed in simulation up to 125 km.

C. Design rule

Solving $P_{\text{lost}} \le 1\%$ for $T_{\text{seg}} = 1$ s over the pilot spacing n_p gives the smallest pilot share $1/n_p$ that is safe. Table II lists it for three oscillator qualities; the default 1% suffices up to about 120 km for the stable oscillator, while a poor oscillator at 100 km needs 3.2%.

D. Accuracy of acquisition

For a line fitted to n equally spaced measurements of noise σ, the Cramér–Rao bound [23] on the standard deviation of the fitted value at the last point is

$$\sigma_{\text{CRB}} = \sigma \sqrt{\frac{4n - 2}{n(n+1)}} \approx \frac{2\sigma}{\sqrt{n}}. \tag{14}$$

For $\sigma = 0.35$ ns and $n = 64$ this is 86 ps. Our acquisition (Fig. 2c) reaches 96 ps, 11% above the bound; at $n = 32$ it is 7% above and at $n = 16$ 21% above. With only 4–8 detections, dark counts occasionally mislead the peak search.

V. Timing-Layer Security

A. Detector mismatch and time shifts

The two detectors of Bob (bit 0 and 1) have gates centred at $c_0 = -\delta/2$ and $c_1 = +\delta/2$ around the synchronized centre, where δ is the mismatch [24]. A photon arriving at relative time s is detected by detector b with efficiency $\eta_b(s) = \Phi\big(\frac{c_b+W/2-s}{\sigma}\big) - \Phi\big(\frac{c_b-W/2-s}{\sigma}\big)$. If a spy controls the arrival shift s, a click reveals the bit with posterior $\eta_1/(\eta_0+\eta_1)$, and the information is $I(s) = 1-h\big(\eta_1/(\eta_0+\eta_1)\big)$ bits per sifted bit, without raising the QBER [25], [26].

B. Delaying a classical reference, and in-band immunity

If timing comes from a classical reference, a delay by s of that reference shifts Bob's gates relative to the photons by s. In-band synchronization derives timing from the photons, so a uniform delay is estimated as part of $e(t)$ and the relative shift remains zero: $I = 0$.

C. Random time shifts and a variance monitor

If the spy shifts each photon by $\pm s$ at random, the tracker cannot absorb it, but the residual variance grows from σ^2 to about $\sigma^2 + s^2$. For n residuals the sample variance is approximately normal with mean σ^2 and variance $(\mu_4-\sigma^4)/n$ (μ_4 the fourth moment); the monitor raises an alarm when it exceeds the threshold for a false-alarm rate of 10^{-3}. The moments are estimated from a large sample of detected residuals.

D. Leakage from synchronizing on key detections

The time tags and the tracker state remain inside Bob's receiver, and only bases and tested bits are announced, so the tracker adds no public channel. In an ideal receiver the arrival time is independent of the bit. The only coupling is the mismatch: a tracker that pools the tags of both detectors is biased by

$$b = \frac{\delta}{2}\,(p_1 - p_0), \tag{15}$$

where $p_1 - p_0$ is the click imbalance between the detectors, which for random bits is about $1/\sqrt{N}$ for N detected clicks (10^{-2} for 10^4 clicks, 10^{-3} for 10^6). For $\delta = 0.8$ ns and an imbalance of 10^{-3} the bias is 0.4 ps and $I \approx 9 \times 10^{-9}$ bit; even for an extreme imbalance of 5% the bias is 20 ps and $I \approx 2 \times 10^{-5}$ bit. Delays and random shifts imposed by the spy are handled as above, with the monitor now having up to 10^6 detections instead of 10^2–10^4 pilot clicks. This is a bounded argument with numbers, not a composable security proof of the whole synchronization layer; active injection of false synchronization clicks is not analysed.

VI. Numerical Results

A. Setup

All curves come from the simulator described in Sec. II. Each point is the average (or, for timing errors near lock loss, the median) of 50–1 000 independent blocks of 10^8–10^{10} slots, so that a point rests on 10^{10}–10^{12} simulated pulses and a figure on 10^{11}–3×10^{14}. Error bars are 95% confidence intervals of the mean, and Wilson intervals for proportions; the median relative half-width is 0.002% for the QBER, 0.1–0.6% for key rates and 1.1% for the timing error of Fig. 4. All random numbers derive from fixed seeds, and the code and the data behind every marker are provided with this paper. The sync-phase timing error is the RMS of the gate error over the key part of the 1 s segment. We say a run *loses lock* if more than 1% of the key time has $|d| > W/2 + 3\sigma = 2.05$ ns.

B. Validation

Figure 2 confirms that the simulation reproduces the closed-form QBER, the click statistics, and, in panel (c), that the acquisition error approaches the Cramér–Rao bound within 7–11% for 32–64 detections.

C. Robustness to clock impairments

Figure 3 shows the timing capture at 25 km against initial offset, drift and jitter. Without synchronization the capture is below 1% except at zero drift. A classical reference with 0.5 ns residual loses 8–12 percentage points (88–92%), batch re-synchronization keeps 95–97%, and preamble-only QCS keeps 6–10% because its extrapolated drift error grows over the block. Kalman tracking on pilots keeps 99.5% across 0–20 ns offset and 0–200 ppm drift, within 0.05% of perfect synchronization. At large jitter all methods converge to the limit set by W/σ.

D. Kalman tracking versus published estimators

Figure 4 compares the four QCS estimators of Sec. III for a stable and a poor oscillator, and Table III summarizes it. On the same detections, Kalman tracking on key clicks has a 3–5 times lower RMS gate error than the tuned histogram method (13–39 ps against 40–201 ps for the stable oscillator from 25 to 125 km, 28–84 ps against 81–420 ps for the poor one) and more than 10 times lower than batch re-synchronization (244–818 ps, stable oscillator). It does not lose lock up to 125 km in either case (8% of the runs lose lock at 150 km with the poor oscillator). Kalman tracking on pilots is 3–10 times more accurate than batch re-synchronization on the same pilots from 10 to 100 km. The batch method loses lock at 125 km with the stable oscillator and from 75 km with the poor one; with the poor oscillator Kalman tracking on pilots holds up to 75 km, keeps 71% of the ideal key at 100 km, and fails at 125 km. With key detections the tracker is updated every 1–50 μs instead of every 11 μs–3.3 ms, which explains its advantage over the pilot version.

E. Key rate against distance

Figure 5 shows the finite-key rate against distance. All methods coincide on the logarithmic scale up to 75 km; the differences are in the relative gap to perfect synchronization (finite key). At 100 km the key rate relative to perfect synchronization is 99.9% for Kalman on key clicks, 98.0% for Kalman on pilots, 96.4% for the tuned histogram method, 93.3% for batch re-synchronization and 87.8% for the classical reference; the mean QBER of the histogram method is 4.3%

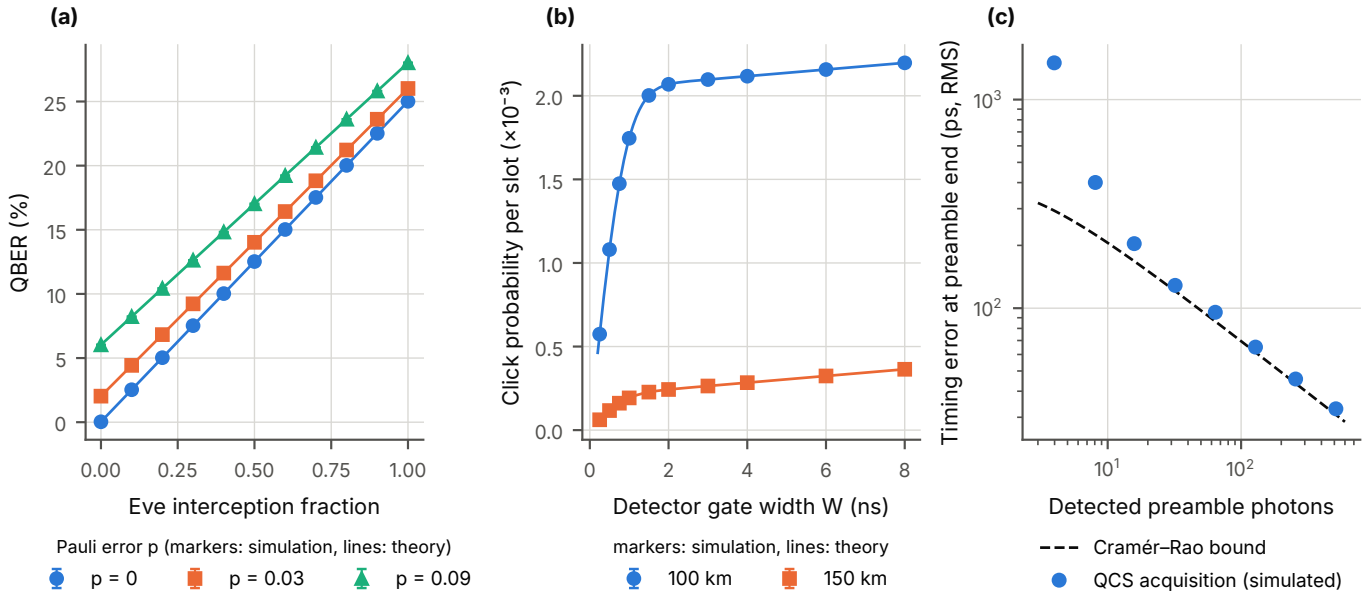


Fig. 2. Model validation. (a) QBER against the spy's interception fraction for three channel noise levels; markers: simulation, lines: closed form. (b) Click probability per slot against gate width at 100 and 150 km; lines: Eq. (1) with Eq. (4). (c) Timing error after the preamble against the number of detected photons, with the Cramér–Rao bound.

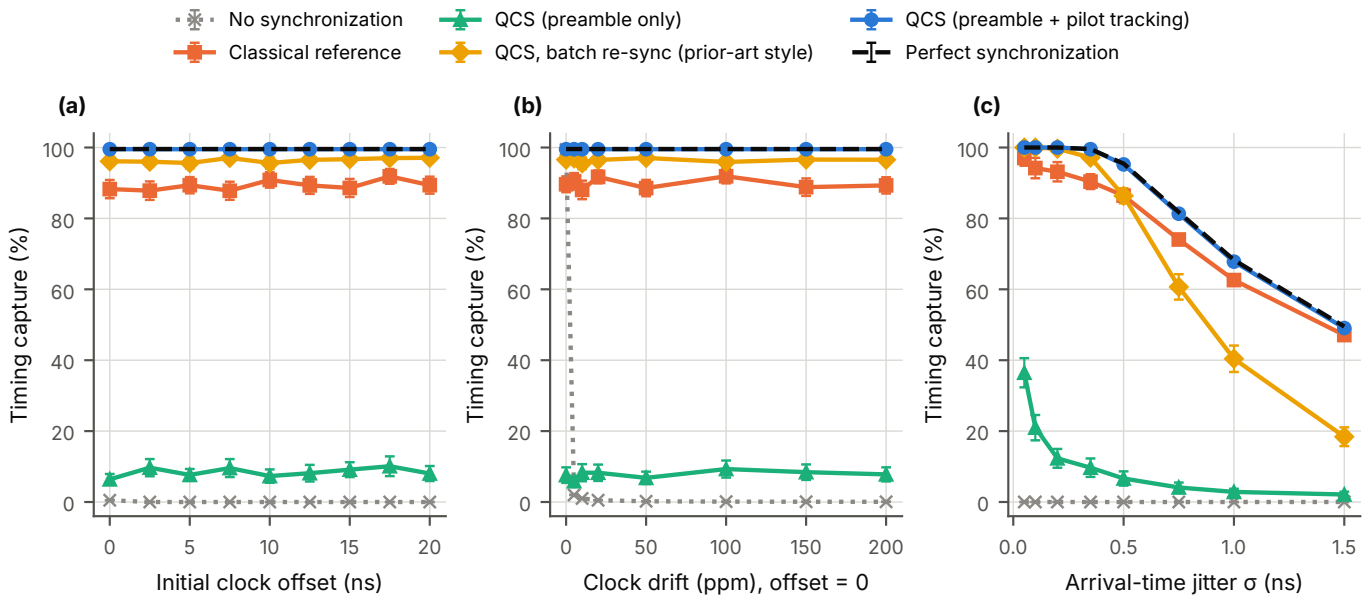


Fig. 3. Timing capture at 25 km against (a) initial offset, (b) drift and (c) jitter for no synchronization, classical reference, preamble-only QCS, batch re-synchronization, Kalman tracking on pilots, and perfect synchronization.

against 2.9% for the others, because it includes the 3% of its runs that lose lock. Kalman tracking on key clicks exceeds the classical reference by 7.5% at 25 km, 12% at 50 km, 15% at 75 km and 14% at 100 km. At 125 km only the Kalman variants give any key; the rate there is a few hundred bits per block, so the percentages are noisy (58% of perfect here, 45% in Fig. 4), and nothing is extractable at 150 km with blocks of 10^{10} slots. The infinite-key limit for this setup ends at 143 km. Preamble-only QCS gives essentially no key at any distance.

F. Finite-size behaviour

Figure 6 shows the key rate against the key block size from 10^8 to 10^{11} slots. At 25 km, Kalman tracking on pilots stays within 1–2% of perfect synchronization for every size, and the classical reference is 9–16% lower. At 100 km no key is possible below 3×10^9 slots, and above it Kalman tracking is within 3% of perfect synchronization. Both approach the asymptotic limit slowly, which is why the composable bound matters for the comparison.

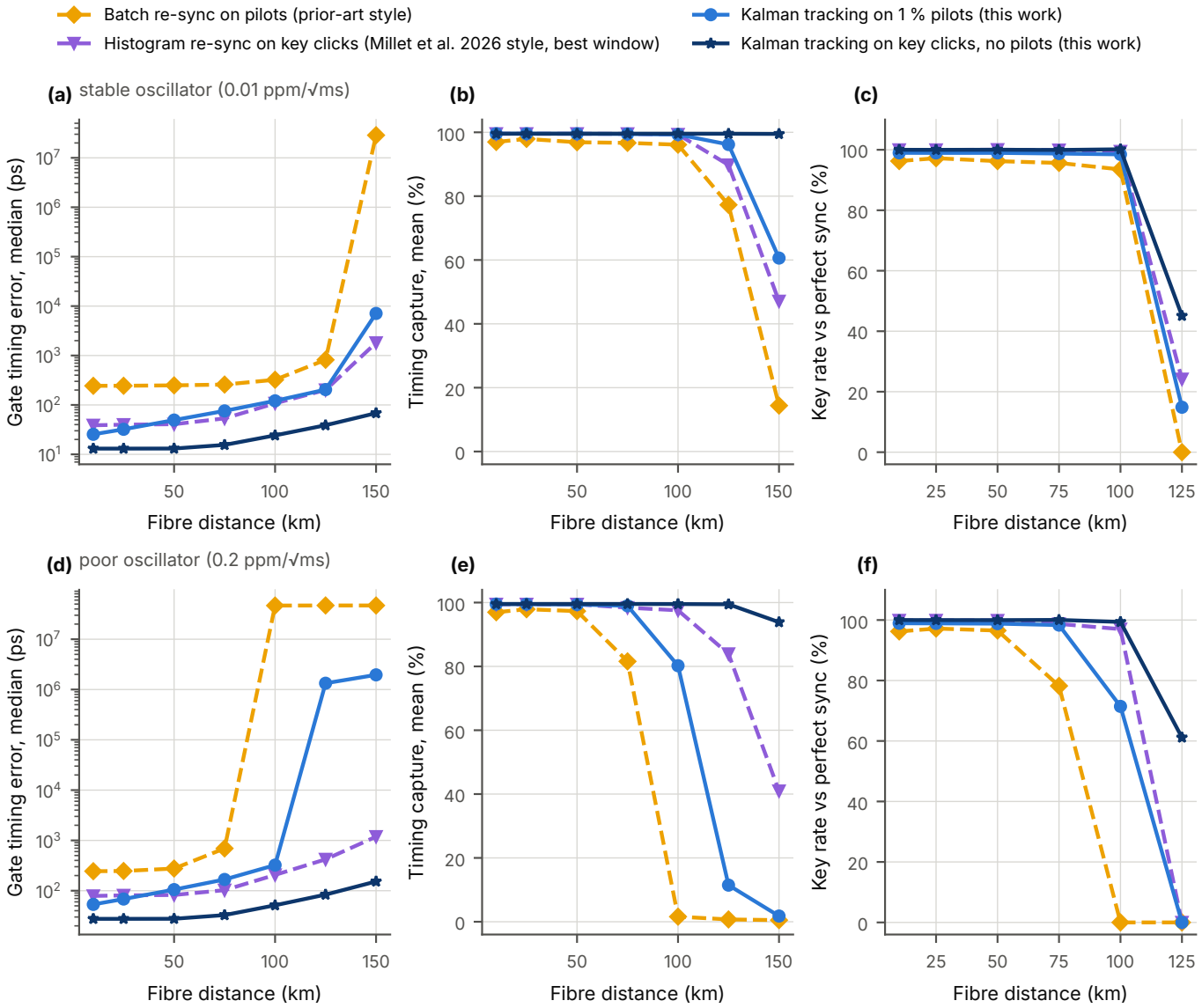


Fig. 4. Kalman tracking versus published-style estimators for a stable (top) and a poor (bottom) oscillator: (a, d) median RMS gate error, (b, e) timing capture, (c, f) key rate relative to perfect synchronization. Histogram re-synchronization is tuned to its best duration at every point.

G. Theory against simulation

Figure 7 compares (11) and (13) with the simulation. The mean-square error formula reproduces the simulated RMS error for the pilot tracker to within a few percent where the tracker is locked (31.7 against 32.1 ps at 25 km and 117 against 121 ps at 100 km for the stable oscillator; 67 against 68 ps and 286 against 323 ps for the poor one), and for the key-detection tracker, with Δ replaced by the mean gap between processed clicks, within 1–4% up to 100 km and 8–9% at 125 km; at 150 km the simulation is 23–27% above the theory. The lock-loss formula places the 50% point at 132 km (stable) and 90 km (poor); the simulation shows 3% of runs lost at 125 km and 76% at 150 km (stable), and 41% at 100 km and 100% at 125 km (poor), so the boundary is predicted to within one step of the 25 km grid.

H. Fairness checks

Figure 8 tests whether the advantage depends on a poorly tuned baseline. (a) For the poor oscillator, the best batch size (64 clicks, 207 ps at 25 km; 16 clicks, 1.3 ns at 100 km) is still 3 and 4 times worse than Kalman tracking on the same pilots (69 and 287 ps). (b) A classical reference with rms residual below about 0.2 ns achieves at least 99% of the perfect-synchronization key rate and matches in-band tracking; at 0.5 ns it gives 83–88% and at 1 ns 56–67%. The benefit of in-band synchronization over a classical link is therefore not accuracy against a high-quality link, but the removal of the separate link and its attack surface. (c) For the histogram method on the same key detections, the best duration (300 μs: 80 ps at 25 km and 206 ps at 100 km) is still 3 and 4 times worse than Kalman tracking (28 and 51 ps); short windows are too noisy at long range and long windows let the oscillator wander between corrections.

TABLE III
MEDIAN RMS GATE ERROR (PS) AND KEY RATE RELATIVE TO PERFECT SYNCHRONIZATION. "LOST": MOST RUNS LOSE LOCK. STABLE / POOR OSCILLATOR.

Method	25 km	100 km	125 km	Key at 100 / 125 k
Stable oscillator				
Batch (pilots)	244	323	818	94% / 0%
Histogram (key clicks)	40	108	201	99.5% / 24%
Kalman (pilots)	32	121	204	98.5% / 15%
Kalman (key clicks)	**13**	**24**	**39**	**100% / 45%**
Poor oscillator				
Batch (pilots)	247	lost	lost	0% / 0%
Histogram (key clicks)	81	206	420	97% / 0%
Kalman (pilots)	68	323	lost	71% / 0%
Kalman (key clicks)	**28**	**51**	**84**	**99% / 61%**

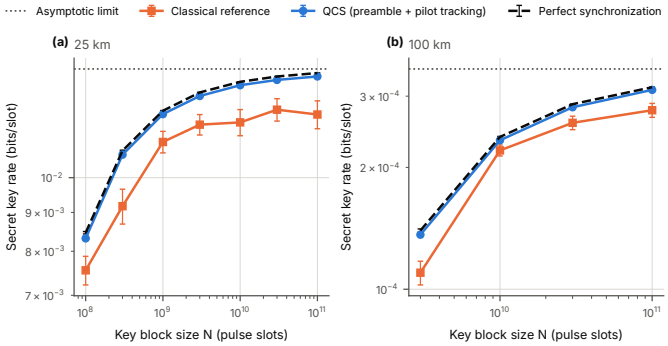


Fig. 6. Secret key rate against key block size at (a) 25 km and (b) 100 km for the classical reference, Kalman tracking on pilots and perfect synchronization; dotted: asymptotic limit.

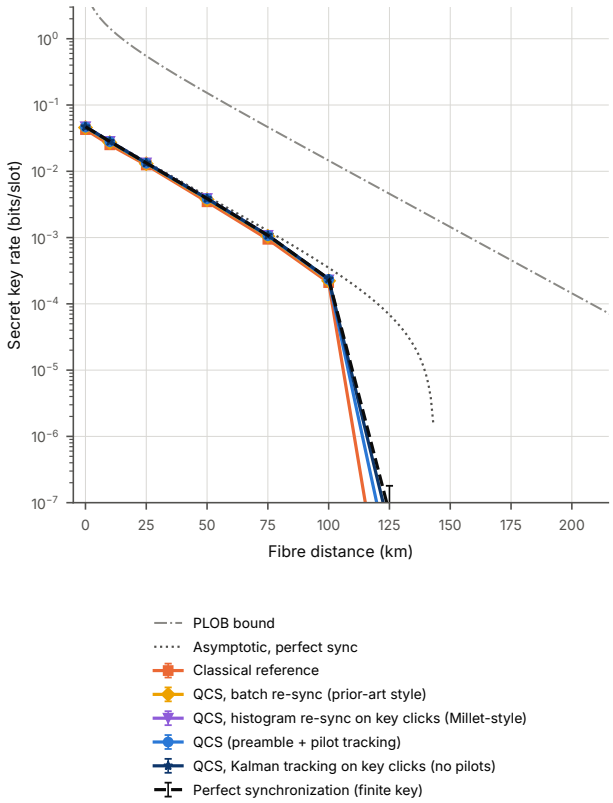


Fig. 5. Finite-key secret key rate against distance for the classical reference, batch, histogram (tuned) and Kalman methods, and perfect synchronization, with the asymptotic perfect-synchronization rate and the PLOB bound.

I. Timing attacks

Figure 9 shows that delaying a classical reference leaks information without a QBER signature: for 0.8 ns mismatch a 1 ns delay gives $I = 0.45$ bit per sifted bit, for 0.4 ns mismatch 0.14 bit, for 0.2 ns mismatch 0.04 bit, while Bob's click rate falls to about half. In-band synchronization absorbs the delay ($I = 0$). A random shift of $\pm s$ is flagged with probability above 90% at a false-alarm rate of 10^{-3} for $s = 0.325$ ns with 100 detections, 0.175 ns with 1 000, and 0.10 ns with 10 000; at these shifts the spy's information is at most 0.0013 bit (0.4 ns mismatch).

VII. Discussion and Limitations

Practicality. The tracker has two state variables and one multiplication-level update per detection or group of detections, and it needs only the time tags that the receiver already records. The pilot version needs the sender to insert sync pulses; the key-detection version needs no change at the sender beyond the preamble, which is the reason we regard it as the preferred variant.

Limitations. (i) The study is simulation only, while Millet *et al.* validated their method on commercial equipment; their method is our reimplementation from the paper, tuned at every point, and implementation details may differ. (ii) The key block statistics assume stationary synchronization after a 1 s timing simulation. (iii) The lock-loss model is derived for the pilot tracker; the key-detection tracker has a matching error formula but no lock-loss formula, and a lost lock is not recovered in our model. (iv) The process noise q equals the true oscillator wander in our simulations; the sensitivity to a wrong q has not been studied. (v) The security argument of Sec. V is a bounded argument, not a composable proof, and active injection of false clicks is not analysed. (vi) One detector model, a fixed 2 ns gate, and a fibre channel with constant loss were used. (vii) A classical timing link with sub-0.2 ns residual matches the tracker's key rate.

Future work. A composable security proof including the synchronization layer; lock-loss theory and re-acquisition for the key-detection tracker; adaptive estimation of q; longer timing simulations; and validation on recorded time tags from a QKD system.

VIII. Conclusion

Treating clock recovery in photon-counting QKD as recursive state estimation yields a tracker that updates after every detection, rejects dark counts by innovation gating, and needs no extra photons when run on the key detections. Its timing error and lock-loss probability follow from closed-form expressions that agree with simulation, and the minimum pilot share for a given distance and oscillator is obtained from them. Against published batch and histogram estimators, both tuned to their best settings, it lowers the RMS gate error by factors of 3–5 and more than 10 respectively over 25–125 km, and

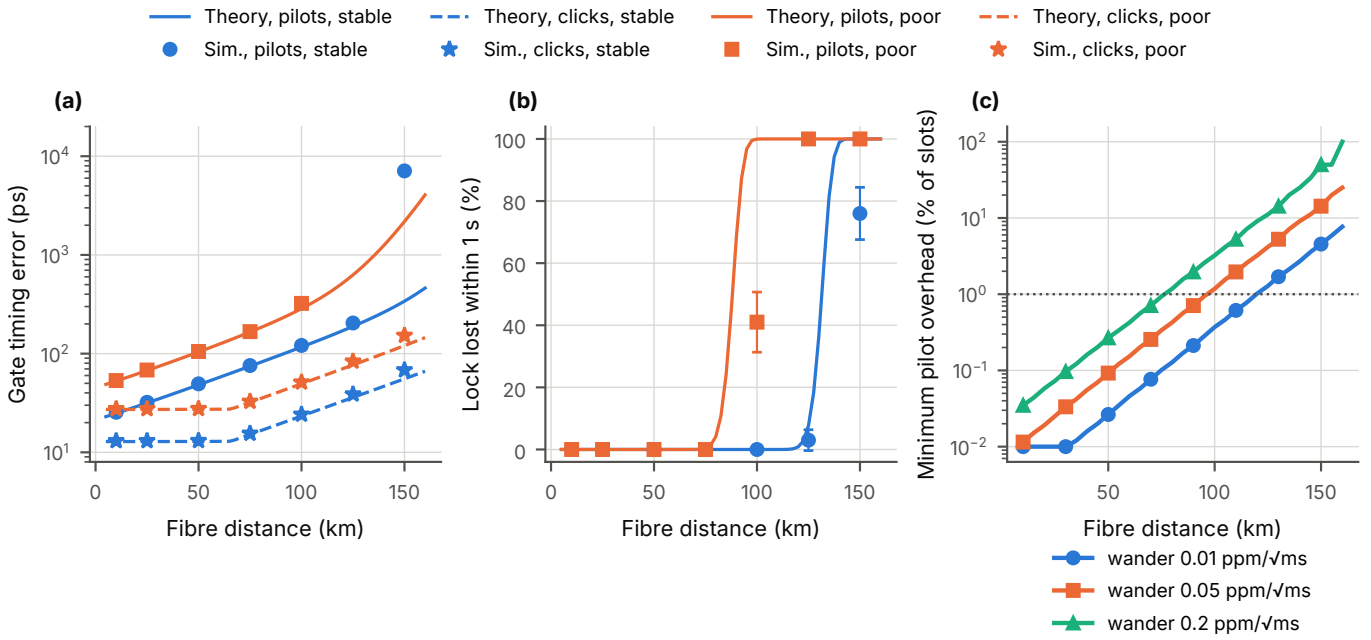


Fig. 7. Analytical model against simulation. (a) RMS gate error for pilots (solid) and key clicks (dashed); (b) probability of losing lock within 1 s for the pilot tracker; (c) minimum pilot share against distance for three oscillator qualities.

retains a substantial share of the ideal key rate at 125 km. A classical reference is competitive only below about 0.2 ns of error and can be attacked through detector mismatch, which in-band synchronization absorbs. Experimental validation and a composable security proof for the synchronization layer remain open.

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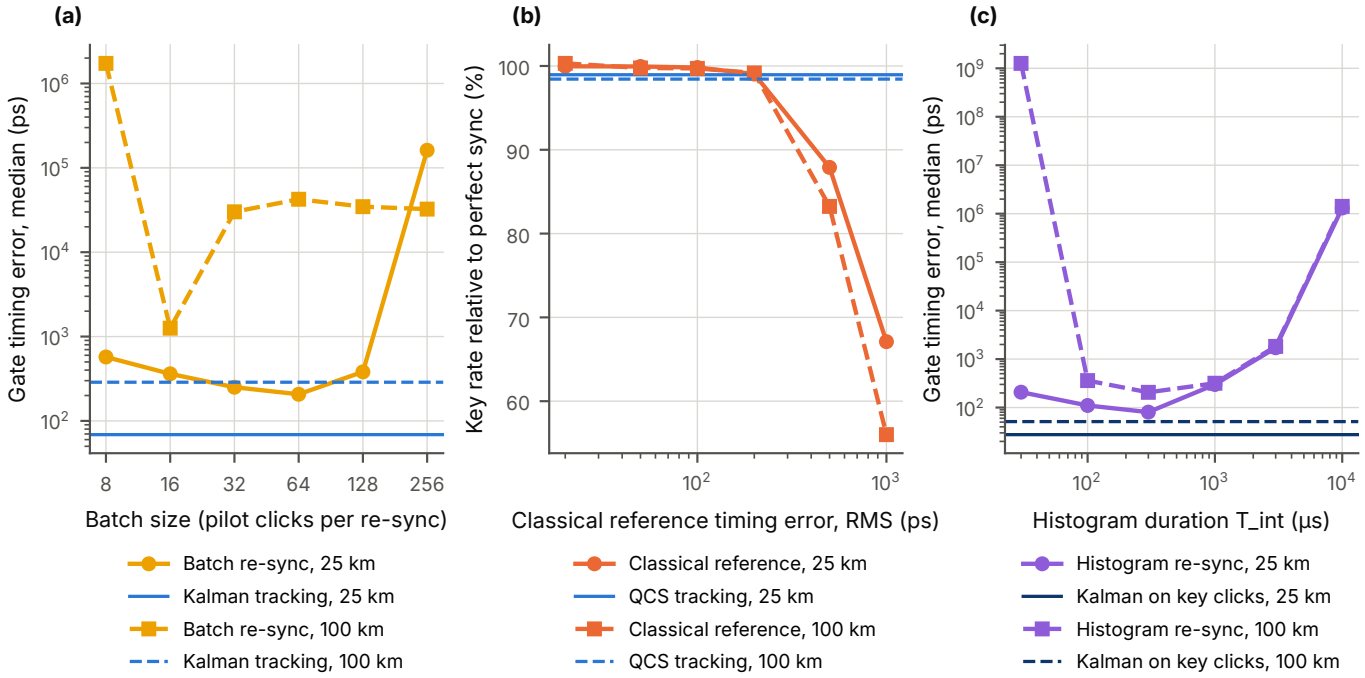


Fig. 8. Fairness checks. (a) Batch size of the batch baseline; (b) quality of the classical reference; (c) histogram duration of the histogram baseline, each against the Kalman tracker (horizontal lines).

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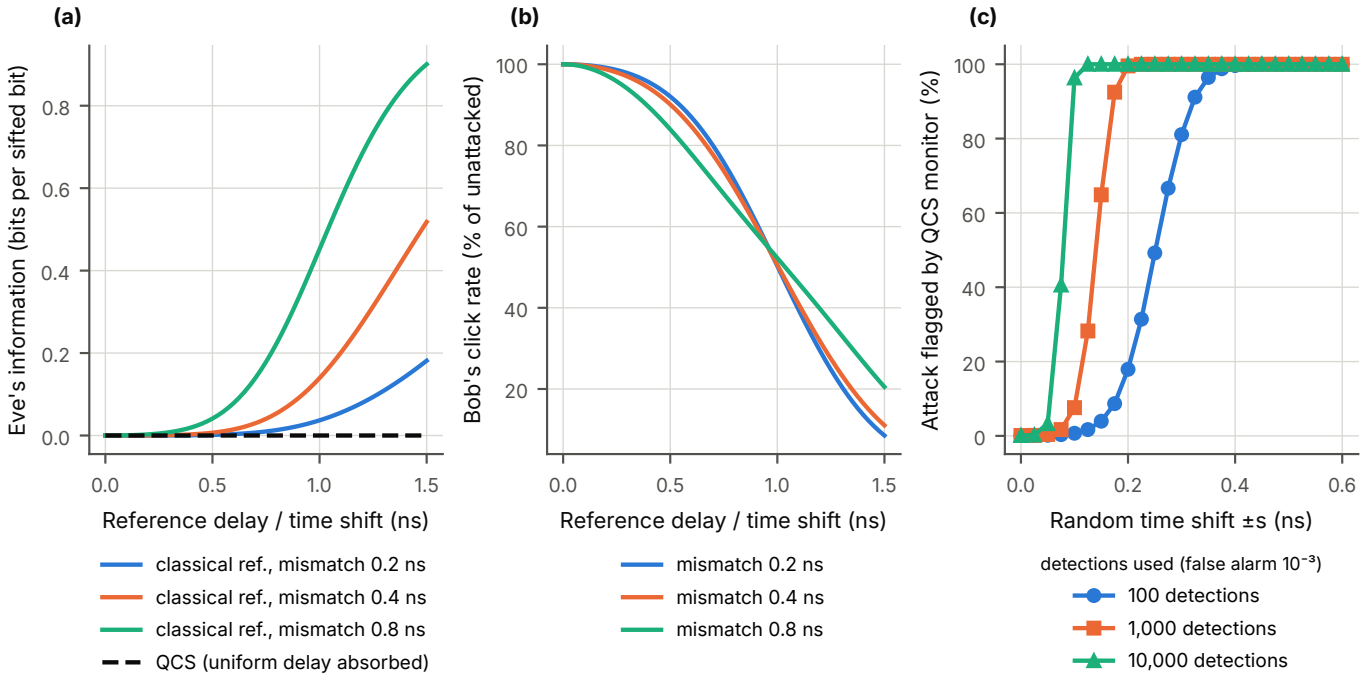


Fig. 9. Timing attacks. (a) Information per sifted bit gained by delaying a classical reference for three mismatches; the dashed line is in-band synchronization. (b) Bob's relative click rate under the delay. (c) Detection probability of the variance monitor against a random $\pm s$ shift for 100, 1 000 and 10 000 detections.